Chimpanzee reference for f4 ratio

2025-12-02

```
res_dir <- "../results/s06_a_chimpanzee_ref/"
```

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Use liftOver to lift hg19 snps to PanTro3.0

Aligned using overchain https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro5.over.chain.gz (2017-05-04) to Pan_tro_3.0 (GCA_000001515.5) from UCSC Genome Browser (Kent et al., 2002).

NOTE: chainfile https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro3.over.chain.gz (2011-02-22) aligns to the 2011 assembly of PanTro3 (GCA_000001515.3) which is older than the 2016 assembly (GCA_000001515.5) used by Ensembl REST API below. This is noted at https://hgdownload.soe.ucsc.edu/goldenPath/panTro5/bigZips/ which states 'This directory contains the May 2016 (Pan_tro 3.0/panTro5) assembly of the chimp genome (panTro5, Chimpanzee Sequencing and Analysis Consortium), as well as repeat annotations and GenBank sequences.' Therefore we use the newer PanTro5 chainfile in the bigsnpr command.

```
bim <- show_loci(mad_bor_g1000_sum)[, c(
  "chr_int", "name", "genetic_dist",
  "position", "allele_ref", "allele_alt"
)]

names(bim) <- c("chr", "name", "genetic_dist", "pos", "a1", "a0")

pan_bim <- snp_modifyBuild(
  info_snp = bim,
  liftOver = "/usr/local/bin/liftOver",
  from = "hg19",
  to = "PanTro5",
  check_reverse = FALSE,
  local_chain = NULL,
  base_url = "https://hgdownload.soe.ucsc.edu/goldenPath/"
)
```

```
## 17765 variants have not been mapped.
```

```
pan_bim <- na.omit(pan_bim)
```

Get chimp reference base for each SNP

Ensembl REST endpoint found here: https://rest.ensembl.org/documentation/info/sequence_region Ensembl uses Pan_tro_3.0 (GCA_000001515.5) 2016/05/03 here https://www.ensembl.org/Pan_troglodytes/Info/Annotation and here https://www.ensembl.org/Pan_troglodytes/Info/Index

```
get_chimp_base <- function(chr, pos, sleep_time = 0.15) {
  # Build region string (example: "1:123456-123456")
  region <- paste0(chr, ":", pos, "-", pos)

  # Ensembl species name for chimp
  species <- "pan_troglodytes"

  # Use the main Ensembl REST API
  url <- paste0(
    "https://rest.ensembl.org/sequence/region/",
    species, "/", region,
    "?content-type=text/plain"
  )

  # Pause briefly to avoid hitting rate limits
  Sys.sleep(sleep_time)

  # Query the API
  r <- GET(url)

  # Check for failure
  if (status_code(r) != 200) {
    warning(paste("Query failed for", chr, pos))
    return(NA)
  }

  # Return the sequence (single base for SNP)
  return(content(r, "text", encoding = "UTF-8"))
}
```

```
chimp_rds <- "../results/s06_a_chimpanzee_ref/mad_bor_g1000_sum_snps_with_chimp.rds"
if (file.exists(chimp_rds)) {
  pan_bim <- readRDS(chimp_rds)
} else {
  tictoc::tic()

  pan_bim$chimp_base <- mcmapply(
    get_chimp_base,
    chr = pan_bim$chr,
    pos = pan_bim$pos,
    mc.cores = 4
  )
  tictoc::toc()
  # Save chimp reference data
  colnames(pan_bim) <- c(
    "chr_int", "name", "genetic_dist", "position",
    "allele_ref", "allele_alt", "chimp_ref_base"
```

```
  )
  saveRDS(pan_bim, chimp_rds)
}
```

Clean up chimp reference data

```
# remove NAs
pan_bim_no_na <- na.omit(pan_bim)
pan_bim$triallelic <- NA
remove_triallelic <- function(pan_bim) {
  three_alleles <- c(
    pan_bim["allele_ref"],
    pan_bim["allele_alt"],
    pan_bim["chimp_ref_base"]
  )
  if (length(unique(three_alleles)) == 3) {
    pan_bim["triallelic"] <- TRUE
  } else {
    pan_bim["triallelic"] <- FALSE
  }
}

pan_bim_no_na$triallelic <- apply(pan_bim_no_na, 1, remove_triallelic)

# filter out triallelics
pan_bim_filtered <- pan_bim_no_na[pan_bim_no_na$triallelic == FALSE, ]
```

```
nrow(pan_bim_filtered)
```

```
## [1] 180431
```

Subset original hg19 gt to only those in chimp reference

```
mad_bor_g1000_sum_chimp <- mad_bor_g1000_sum %>%
  select_loci_if(loci_names(genotypes) %in% pan_bim_filtered$name)
```

Save

```
chimp_gt <-
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_chimp.gt"
if (file.exists(chimp_gt)) {
  file.remove(chimp_gt)
}
```

```
## [1] TRUE
```

```
gt_save(
  mad_bor_g1000_sum_chimp,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_chimp.gt"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_chimp.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_chimp.gt')

## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_chimp.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
```

Create a gen_tibble of the chimp reference alleles

```
if (file.exists("../data/intermediate/pan_tro_liftover/chimp_ref.gt")) {
  file.remove("../data/intermediate/pan_tro_liftover/chimp_ref.gt")
  file.remove("../data/intermediate/pan_tro_liftover/chimp_ref.bk")
  file.remove("../data/intermediate/pan_tro_liftover/chimp_ref.rds")
}
```

```
## [1] TRUE
```

```
loci <- show_loci(mad_bor_g1000_sum_chimp)
indiv_meta <- data.frame(id = "chimp_ref", population = "chimp_ref")
genotypes <- matrix(rep(1, nrow(loci)), nrow = 1, ncol = nrow(loci))

chimp <- gen_tibble(
  loci = loci,
  indiv_meta = indiv_meta,
  x = genotypes,
  backingfile = "../data/intermediate/pan_tro_liftover/chimp_ref"
)
```

```
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/pan_tro_liftove

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/pan_tro_liftover/chi
```

```
## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/pan_tro_lif

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/pan_tro_liftover/chimp_re
```
